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SUPERSEDING
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July 7, 1959 and
Fed. Spec. DD–F–776a
June 30, 1952

FEDERAL SPECIFICATION

FUNNEL, FILTERING, LABORATORY, GLASS, BUCHNER TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—The item covered by this specification shall be a Buchner type glass filtering funnel, suitable for the filtration of fine precipitates and intravenous solutions and general filtration.

1.2 Classification.

1.2.1 *Classes.*—Glass filtering funnels covered by this specification shall be of the following classes, as specified (see 6.1):

Class 1. — Fine fritted glass filtering disc.
Class 2.—Medium fritted glass filtering disc.
Class 3.—Very fine fritted glass filtering disc.
Class 4.—Coarse fritted glass filtering disc.

1.2.2 *Capacities.*—Glass filtering funnels covered by this specification shall be of the following nominal capacities, as specified (see 6.1):

Nominal capacity (milliliters)	
2	
15	
30	
60	
150	
350	(Classes 1, 2, and 4)
600	(Classes 1, 2, and 4)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

DD–G–541—Glass (Laboratory).
UU–P–271—Paper, Wrapping, Waterproofed Kraft.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.
PPP–C–843—Cushioning Material, Cellulosic.
PPP–P–291 — Paperboard, Wrapping, Cushioning.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 626—Glassware; Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Stand-

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ards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–A–140 — Adhesive, Water-Resistant, Waterproof Barrier-Material.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.—The following publication forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Material Standard:

E128–57T — Maximum Pore Diameter and Permeability of Rigid Porous Filters for Laboratory Use.

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material.—The funnels shall be made of borosilicate glass conforming to the requirements of type I, class A of Federal Specification DD–G–541.

3.2 Dimensions. — The dimensions shall conform to table I.

TABLE I.—*Dimensions*

Nominal capacity	Diameter of disc	Height above disc	Length of stem	Outside diameter of stem
Milliliter	*Millimeter*	*Millimeter*	*Millimeter*	*Millimeter*
2	10±0.60	30±5	30±5	7±0.55
15	20±0.85	45±5	70±5	8±0.60
30	30±1.20	45±5	75±5	8±0.60
60	40±1.40	50±5	75±5	10±0.60
150	60±1.80	50±5	75±5	15±0.65
350	80±2.15	75±5	75±5	20±0.85
600	90±2.30	90±5	100±5	20±0.85

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3.3 Pore size.—Maximum pore size shall be as follows and shall be determined as specified in 4.4.1:

Fine—5.5 (maximum) microns.
Medium—15.0 (maximum) microns.
Very fine—2.5 (maximum) microns.
Coarse—60.0 (maximum) microns.

3.4 Annealing.—The funnel shall show no ring strain, and if any longitudinal strain is present, it shall be faint and highly diffused when tested as specified in 4.4.2.

3.5 Reinforcement.—The top edge of the funnel, except the 2 ml. size, shall have a reinforcing bead.

3.6 Identification marking.—Each funnel shall be permanently and legibly marked with the name or registered trademark of the manufacturer. In addition, each funnel shall be marked with the words "fine", "medium", "very fine", or "coarse", or other designation to indicate porosity.

3.7 Workmanship.—The funnels shall be substantially free from marked irregularities, seeds, blisters, bubbles, and other imperfections which would mar the appearance or impair the efficiency of the funnels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection.—The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.—Unless otherwise specified, not less than three funnels of each class and capacity shall be taken at random from each lot of 100 or fraction thereof delivered, for inspection and tests.

4.3 Inspection.—Samples shall be inspected for compliance with the requirements of this specification.

4.4 Tests.

4.4.1 *Pore size test.* — Funnels shall be tested for pore size in accordance with ASTM Method E128–57T.

4.4.2 *Annealing.*—Funnels shall be viewed in a polariscope to determine compliance with method No. 420 of Federal Test Method Standard No. 626.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Each funnel shall be wrapped in a corrugated fiberboard sleeve of suitable size and design. The fiberboard shall conform to Federal Specification PPP–P–291.

5.1.1.1 *Intermediate package.*—The quantity of funnels specified in table II, wrapped as specified in 5.1.1, shall be packaged in a carton of appropriate size conforming to the requirements of Federal Specification PPP–B–636, type I, class 1 or 2, style RSC. Each carton shall be adequately secured to prevent accidental opening. Cartons shall be designed with suitable interior fittings, or shall contain sufficient cushioning material conforming to Federal Specification PPP–C–843 to insure against shifting of, or damage to, the contents.

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TABLE II.—*Packaging quantities*

Nominal capacity	Quantity of funnels to be packaged in each intermediate carton	Quantity of intermediate cartons to be overpacked in each shipping container	Total number of funnels in each shipping container
Milliliter			
2	9	16	144
15	6	16	96
30	6	16	96
60 (Classes 1, 2, and 4)	6	16	96
60 (Class 3)	4	24	96
150 (Classes 1, 2, and 4)	4	8	32
150 (Class 3)	2	16	32
350	3	8	24
600	3	6	18

5.1.2 *Level C.*—Unless otherwise specified, funnels shall be packaged in accordance with accepted commercial practice.

5.2 **Packing.**

5.2.1 *Level A.* — The quantity of funnels specified in table II, packaged as specified in 5.1.1.1, shall be overpacked in an exterior container conforming to Federal Specification PPP–B–621, class 2, type 3 load. Closure and strapping shall comply with appendix thereto. Unless otherwise specified, each box shall be lined with a waterproof barrier material conforming to Federal Specification UU–P–271. The caseliner shall be closed and sealed with water-resistant adhesive conforming to Military Specification MIL–A–140. The seam shall be continuous and at least ¾-inch in width.

5.2.2 *Level B.*—The quantity of funnels specified in table II, packaged as specified in 5.1.1.1, shall be overpacked in an exterior container conforming to Federal Specification PPP–B–621, class 1.

5.2.3 *Level C.*—The funnels shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.2.4 *Packing variation permitted, applicable to level A or level B.* — If the required number of units to be shipped is less than the number specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.3 **Marking.**

5.3.1 *Civil agencies.*—In addition to any special marking required in the contract or order, marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* — Interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. **NOTES**

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

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(*b*) Class and capacity required (see 1.2.1 and 1.2.2).

(*c*) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 **Transportation description.**—Transportation description applicable to this item is:

Glassware, chemical laboratory.

Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Glassware, laboratory.

Motor volume minimum weight 18,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C E M O
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